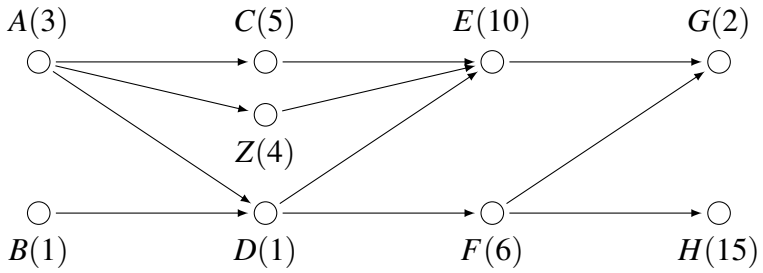
Goal: Learn the Backflow Algorithm and implement the Critical Path Algorithm.

1. Backflow Algorithm
   (a) Introduce an "end" vertex, say X, with a time of $[0]$.
   (b) From X, move back through every vertex assigning it the maximum time to reach vertex X.
2. Apply the **Backflow Algorithm** to the digraph below.



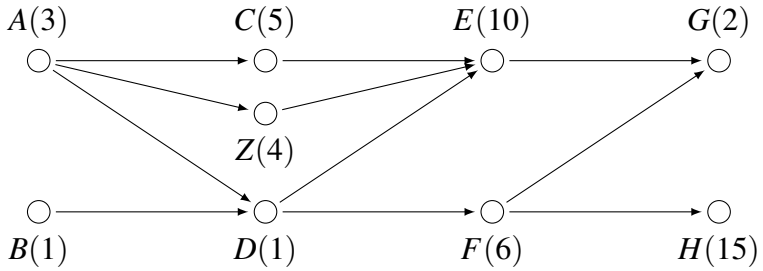
3. Use your work above to answer the questions below.
   (a) Find a critical path and the critical time in the digraph above. What do you observe about the numbers you produced in the Backflow Algorithm?

   (b) Determine the priority list using the **Decreasing Time Algorithm**.

   (c) Determine the priority list by ordering the the vertices according to decreasing **critical time** (the labels from the Backflow Algorithm).

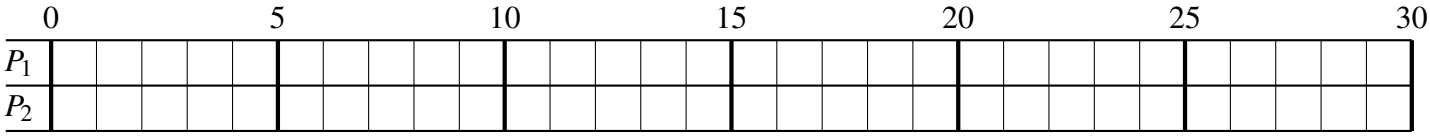
4. Critical Path Algorithm

5. Use your two priority lists from the previous page to construct a schedule using two processors.



(a) decreasing time priority list: _____

time	
done	
ready	



(b) decreasing **critical** time priority list: _____

time	
done	
ready	

